

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2052

Venice Hill 345 kV
February 29, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2052. The Interconnection Customer proposes a 165.8 MW wind facility interconnecting at the Venice Hill 345 kV substation in the West Virginia area. The request was received on April 04, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 493 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$39,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Artesia Country Club / Ross Fork 138 KV	Air Force Base / Artesia Country Club 345 KV L	101.0	1,009	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Artesia Country Club / Ross Fork 138 KV Line	Reconductor line	\$1,229,000	3% (\$39,000)
	Total allocated to project		\$39,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.